Felix Möller

Machine Learning • Computer Vision • LLM & RAG Applications

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Computer scientist (M.Sc., TU Ilmenau, grade 1.3) who trains and evaluates deep learning models in PyTorch and builds LLM-powered applications with FastAPI and React. At the BMW Group, extended a vehicle-fault classifier to tree-structured data (up to +2 percentage points F1) and evaluated a calibration method (MetaCal) with about an order of magnitude lower calibration error than the existing approach. Master's thesis (grade 1.0) on efficient multi-view object recognition: about 88 % less estimated compute on video data at comparable accuracy. Designed and operates a RAG assistant, live at felixmoeller.me, and brings 20 months of full-stack development in an agile team. Looking for a full-time role in ML and AI engineering, data science or computer vision; open to relocating anywhere in Germany or working remotely.

Core Competencies

- **Machine Learning & Deep Learning**: Python, PyTorch, PyTorch Lightning, scikit-learn, Pandas, NumPy, Matplotlib, Plotly; Set Transformer, transfer and teacher-student learning, probability calibration; systematic evaluation with ablation studies, repeated runs and balanced metrics; experiment tracking with Weights & Biases, configuration with Hydra.
- **Computer Vision**: CNNs (ConvNeXt, EfficientNet) with torchvision, multi-view image and video classification, frame and view selection for efficient inference, data augmentation (RandAugment), synthetic image corruptions (blur, noise, occlusion) for robustness, OpenCV.
- **Generative AI & LLM Applications**: Retrieval-Augmented Generation (RAG) on the Claude API (citations, prompt caching, system prompts, model fallback), hybrid lexical and semantic search (BM25 + multilingual embeddings, Reciprocal Rank Fusion), LLM evals, prompt-injection mitigation; agentic coding with Claude Code.
- **Backend & Deployment**: FastAPI, Pydantic, Server-Sent Events; Linux server operations (systemd, Caddy), deploy script with health checks, uptime monitoring, GitHub Actions, Vercel; AWS (S3, SageMaker); Docker (learning it in a microservice side project).
- **Software Engineering & Data**: Java, Spring Boot, JSF/PrimeFaces, JavaScript/React; SQL, PostgreSQL, Snowflake, Streamlit; testing with pytest, JUnit and Selenium; Scrum, Git, Jira.

Professional Experience

- **BMW Group** **Munich, Germany**
  Data Analytics Intern (Machine Learning) *Apr 2025 - Aug 2025*
  AI model classifying root causes of vehicle faults from workshop data. Python, PyTorch, Pandas, SQL, Snowflake, AWS.
  - Extended the fault classifier to tree-structured JSON data with a recursive Set Transformer and PyTorch nested tensors (no padding): up to +2 percentage points F1 in affected use cases
  - Benchmarked calibration methods (BBQ, ENIR, MetaCal) against the existing isotonic regression; implemented MetaCal with about an order of magnitude lower Expected Calibration Error and built a custom MLP calibrator with a differentiable ECE loss
  - Rebuilt the analysts' Streamlit dashboard on Snowflake from scratch: change-aware caching and lightweight row markers removed waits of 5+ seconds after every interaction
  - Prototyped computer vision integration (zero-shot image categorization, per-category EfficientNetV2-S embeddings); recommended stopping after finding only about 1 % of cases had images
  - Presented results to the team with slides, whiteboard and live demos

IBYKUS AG **Erfurt, Germany**
Working Student, Web Development (Full-Stack Java) *Aug 2023 - Mar 2025*
E-government web portal for managing public funding, Scrum team. Java, JSF/PrimeFaces, JavaScript, SQL, Selenium, JUnit.

- Developed frontend and backend features, including sub-projects for the HALM2 agri-environmental funding program (Hesse)
- Led a frontend taskforce: accessible mobile prototype and frontend templates reused by backend developers
- Built a brute-force form-validation test framework that drives the browser via Selenium: 100 % test coverage of all forms in the application
- Wrote unit and integration tests, fixed bugs and onboarded interns

Education

Technische Universität Ilmenau **Ilmenau, Germany**
M.Sc. Computer Science (Informatik), Grade 1.3 (German scale, 1.0 = best) *Apr 2024 - May 2026*
Thesis (grade 1.0): "Learning to Select Informative Views for Efficient Multi-View Object Recognition." Designed a two-stage PyTorch pipeline: a lightweight frame scorer (EfficientNet Lite 0), trained teacher-student style on calibrated classifier confidences, selects the most informative frames for a multi-view CNN (ConvNeXt Tiny). Evaluated on four datasets: about 88 % less estimated compute on video data at comparable balanced accuracy (at most 0.1 pp loss); 97 % recall for corrupted frames.

Technische Universität Ilmenau **Ilmenau, Germany**
B.Sc. Computer Science (Informatik), Grade 1.8 *Oct 2020 - Jun 2024*

Projects

felixmoeller.me
Portfolio Website with AI Assistant *Oct 2026 - Present*
Solo project, built with Claude Code. Python, FastAPI, Claude API, RAG, React, pytest.

- Designed an AI assistant that answers recruiter questions with source citations: hybrid BM25 + multilingual embedding search fused via Reciprocal Rank Fusion, answers streamed via Server-Sent Events
- Secured with an HMAC-signed chat history against prompt injection, rate limits, a daily cost budget and Pydantic input validation; GDPR-compliant (no IP logging); about 100 pytest tests plus LLM evals with trick questions
- Runs the backend on a self-managed Linux server (systemd, Caddy) with a health-checked deploy script and uptime monitoring; bilingual React frontend; CV built as code by GitHub Actions

Languages

- German (native), English (C1, fluent), Chinese (basic, HSK 1).

References

- Available upon request.